Day 138 | Unit 10 | Teacher plan

AASA Applications and Cumulative Review

Connect Coordinates and Geometry Models

Standards: 6.NS.C.8, 6.G.A.3, 6.G.A.1, 6.G.A.4, 6.G.A.2

Learning target: I can connect Coordinates and Geometry Models, and explain and check my reasoning.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Plot both points. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units. On opposite sides of zero, add distances to zero. On the same side, subtract magnitudes. Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths. A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 4, 7, 19, 23

Day 138 | Unit 10 | Lesson

AASA Applications and Cumulative Review

Connect Coordinates and Geometry Models

Standards: 6.NS.C.8, 6.G.A.3, 6.G.A.1, 6.G.A.4, 6.G.A.2

Learning target: I can connect Coordinates and Geometry Models, and explain and check my reasoning.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

01 / Learn

Plot both points. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units. On opposite sides of zero, add distances to zero. On the same side, subtract magnitudes. Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths. A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

Worked example 1

Plot (-3, 5) and (-8, 5). Find their distance.

1. Plot both points.

2. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units.

3. Only one coordinate changes.

4. Same side of 0: subtract magnitudes to get 5.

5. State the result in the form requested: 5 units.

Answer: 5 units

Worked example 2

A prism measures 13/2 m by 2 m by 2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Multiply edges: 13/2 × 2 × 2 = 26.

3. The edge counts are 13, 4, 4, giving 208 cubes.

4. Each cube is 1/8 m^3

5. 208/8 = 26.

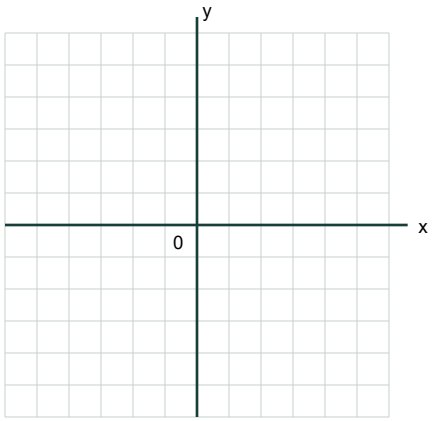
Answer: 26 m^3; 208 cubes

Day 138 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

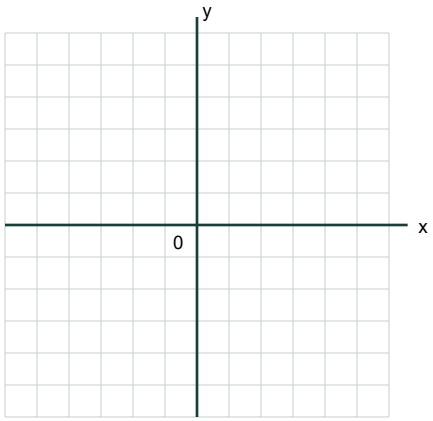
Plot (4, -6) and (4, -10). Find their distance.



Label the scale and units.

02 / Constructed response

Plot and connect (0,0), (12,0), (12,8), (0,8). Find the side lengths and area.



Label the scale and units.

03 / Constructed response

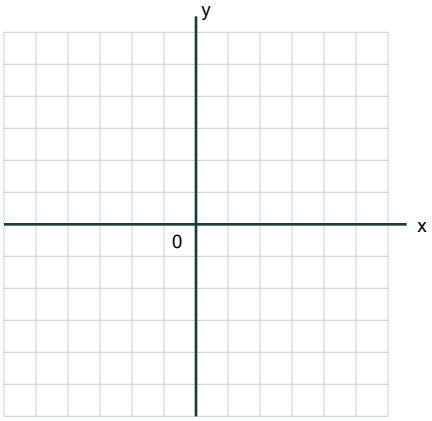
A 5 cm by 5 cm base and 6 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

04 / Constructed response

A rectangular prism has edges 3, 3, and 3 cm. Sketch it and find its volume.

05 / Constructed response

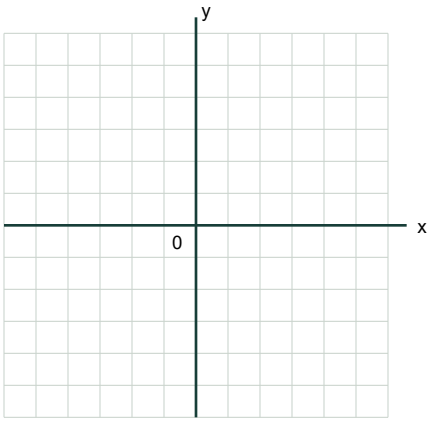
Plot (6, -4) and (6, 5). Find their distance.



Label the scale and units.

06 / Constructed response

Plot and connect (-6,-4), (4,-4), (4,4), (-6,4). Find the side lengths and area.



Label the scale and units.

07 / Constructed response

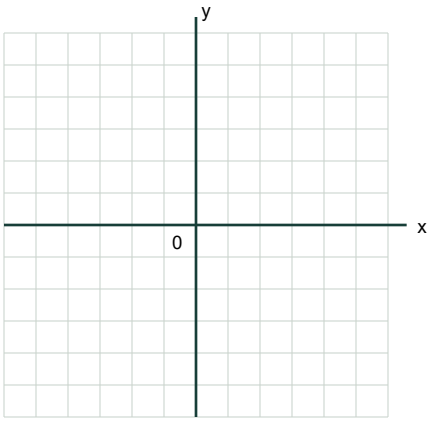
A 4 cm by 2 cm base and 5 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

08 / Numeric entry and model reasoning

A prism measures 5/2 m by 1 m by 5/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Coordinate plotting and numeric entry

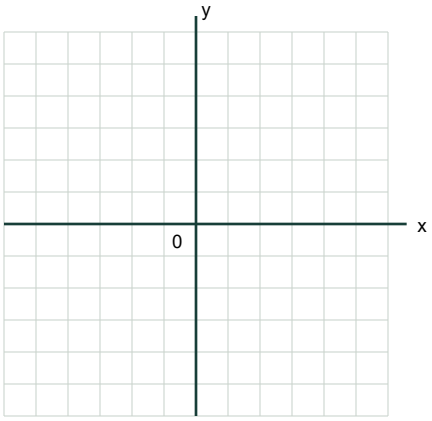
Plot A(-7, -6), B(5, -6), and C(-7, 6). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

10 / Constructed response

Plot and connect (-3,-6), (3,-6), (3,3), (-3,3). Find the side lengths and area. Find the perimeter too.



Label the scale and units.

Day 138 | Unit 10 | Answer key

AASA Applications and Cumulative Review

Connect Coordinates and Geometry Models

Standards: 6.NS.C.8, 6.G.A.3, 6.G.A.1, 6.G.A.4, 6.G.A.2

Learning target: I can connect Coordinates and Geometry Models, and explain and check my reasoning.

1. 4 units

Only one coordinate changes. Same side of 0: subtract magnitudes to get 4.

6.NS.C.8 | Constructed response

2. Sides 12 and 8; area 96 square units

Horizontal length 12 - (0) = 12; vertical length 8 - (0) = 8; area = 12 × 8.

6.G.A.3, 6.G.A.1 | Constructed response

3. 170 cm²

Closed surface area = 2(25+30+30) = 170.

6.G.A.4 | Constructed response

4. 27 cm³

B = 3 × 3 = 9 cm²; V = B × 3 = 27 cm³.

6.G.A.2 | Constructed response

5. 9 units

Only one coordinate changes. Opposite sides of 0: 4 + 5 = 9.

6.NS.C.8 | Constructed response

6. Sides 10 and 8; area 80 square units

Horizontal length 4 - (-6) = 10; vertical length 4 - (-4) = 8; area = 10 × 8.

6.G.A.3, 6.G.A.1 | Constructed response

7. 68 cm²

Closed surface area = 2(8+20+10) = 76. Remove the top area 8: 68.

6.G.A.4 | Constructed response

8. 25/4 m³; 50 cubes

Multiply edges: 5/2 × 1 × 5/2 = 25/4. The edge counts are 5, 2, 5, giving 50 cubes. Each cube is 1/8 m³; 50/8 = 25/4.

6.G.A.2 | Numeric entry and model reasoning

9. AB = 12 units; AC = 12 units

AB is horizontal: 7 + 5 = 12. AC is vertical: 6 + 6 = 12. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

10. Sides 6 and 9; area 54 square units; perimeter 30 units

Horizontal length 3 - (-3) = 6; vertical length 3 - (-6) = 9; area = 6 × 9.

6.G.A.3, 6.G.A.1 | Constructed response